Yilun Wang

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Education

Ph.D. in Economics, North Carolina State University, Raleigh, North Carolina 2020 – 2025

- Dissertation: *"Essays on Asset Pricing with Deep Learning Methods and Large Language Models"*
- Committee: Profs. Mehmet Caner (Advisor), & Ilze Kalnina, & Denis Pelletier, & Zheng Li

M.A. in Economics, Northeastern University, Boston, Massachusetts 2019 – 2020

M.S. in Economics, Chinese University of Hong Kong, Shatin, Hong Kong 2017 – 2018

B.A. in Economics, Southwestern University of Finance and Economics, Chengdu, China 2013 – 2017

Research Fields

Applied Econometrics, Asset Pricing, Language Models, Financial Economics, Machine Learning

Publications

"CVRA: Asset Pricing via the Conditional Variational Recurrent Autoencoder in Asian Market," **Asia-Pacific Financial Markets**, pp.1-45 (May 2026)

Working Papers (Finance Track)

"Conditional Dynamic Latent Factor Modeling for Asset Pricing" Revise & Resubmit at **Journal of Banking & Finance**

- **Abstract**: We develop a dynamic conditional latent factor model for asset pricing that explicitly models the temporal evolution of latent systematic risk. The framework combines a sequential factor learning block, which extracts latent factors from recent return history using a recurrent encoder and one-sided attention, with a characteristic-based beta network that maps firm characteristics into time-varying factor loadings. Using monthly U.S. stock returns from 1980 to 2023 and 64 firm characteristics, we evaluate the model in a strict rolling out-of-sample design against linear, machine learning, and deep learning benchmarks. The proposed model delivers the strongest total and predictive R^2, as well as the highest Sharpe ratios in both long-short and long-only portfolio strategies. Additional evidence shows that the learned latent factors are economically interpretable and capture dynamic information not fully spanned by standard benchmark factors or observed macro-financial variables.

"Opening the Black Box: Residual Neural Information in Return Prediction" Under Review at **Journal of Financial and Quantitative Analysis**

- **Abstract**: This paper opens the black box of neural return prediction from price-volume histories. We decompose each neural forecast into a characteristic-spanned score and a residual-implied score by projecting hidden representations onto standard price-volume characteristics and carrying the decomposition through the prediction head. Neural networks recover familiar trading states, especially when temporal ordering is modeled, but return predictability is concentrated in the residual-implied score. The residual component survives controls, is strongest when arbitrage is limited, and forecasts future trading-state transitions. Deep learning therefore does more than recombine known characteristics nonlinearly; it extracts high-dimensional residual information from price-volume histories.

"News Not Yet in Prices: A Multimodal Factor for Expected Returns" Submitted

- **Abstract**: We study which part of public news is not yet incorporated into stock prices. We combine an MLP representation of firm-level price–volume histories with contextual embeddings of timestamped news and construct a news innovation score that predicts the component of future returns unexplained by price histories and standard firm characteristics. In U.S. equities from 2000 to 2024, the resulting News Innovation Factor earns a higher value-weighted long–short return than benchmarks and remains significant after controlling for established factors and machine-learning forecasts. Predictability is strongest for novel, complex, and weakly attended news and is followed by analyst revisions and realized fundamentals, consistent with gradual information diffusion. The factor expands the mean–variance frontier and reduces pricing errors across anomaly portfolios. Our findings show that prices and news contain complementary information and that multimodal learning can measure public information not yet reflected in market prices.

Working Papers(AI Application Track)

'FedAgent: A Multi-Agent System Framework for Federal Funds Target Rate Prediction" Under Review at 2026 Neurips

- **Abstract**: Decisions of the Federal Open Market Committee (FOMC) regarding the federal funds rate play a central role in shaping financial market conditions and macroeconomic dynamics. However, forecasting the decisions is challenging due to the complicated committee-based nature. Most existing approaches map economic indicators or textual signals directly to policy actions, abstracting away from institutional deliberation. We propose FedAgent, an institution-constrained multi-agent framework that models FOMC interest rate decisions as a structured process of collective reasoning. The framework decomposes policymaking into evidence construction, discrete policy option formation, role-conditioned deliberation, and rule-based aggregation, while explicitly modeling role heterogeneity and asymmetric informational salience. Evaluated on scheduled FOMC meetings from 2020 to mid-2025, FedAgent achieves higher accuracy and greater prediction stability than strong machine learning and LLM-based baselines, while providing interpretable diagnostics of committee disagreement and consensus.

'FusionITT: Dominance-Guided Multimodal Transformer for Financial Time-Series Forecasting" Under Review at 2026 Neurips

- **Abstract**: Financial market prediction relies on heterogeneous signals from price time series, chart images, and textual news. Existing multimodal methods often treat all modalities symmetrically, ignoring the dominant role of time-series dynamics and thus suffering from noisy cross-modal interference. We propose FusionITT, a dominance-guided multimodal Transformer for stock movement direction prediction. FusionITT anchors fusion on time-series representations and introduces a Hyper-Representation Alignment Module (HRAM) to selectively align and filter auxiliary visual and textual information under multi-scale temporal guidance. An asymmetric cross-modal attention mechanism further integrates complementary signals. Experiments against state-of-the-art multimodal models and large language models demonstrate that FusionITT achieves superior performance and robustness in volatile financial environments.

Work in Progress

"Agentic Framework for Alpha Mining"

"Multi-Agent Framework for Quantitative Investment"

"View From Board: Financial Statement Analysis with Large Language Models"

"Relaxation-and-Refinement Policy in Improving Soft Actor-Critic (SAC) Agent for Portfolio Optimization"

Working Experience

Research Fellow, Jinan University, Center of Bid Data and Management 03/2025-Now

Senior Quantitative Researcher, Noah Ark Quant Management LLC 03/2025-Now

Quantitative Algorithm Engineer, Hithink Flush Information Network Co., Ltd 07/2024-03/2025

Teaching Experience

Independent Instructor, NC State University 6 semesters

- Principles of Macroeconomics: Fall 2022, Spring 2023, Summer 2023, Fall 2023, Spring 2024, Fall 2024

Graduate Teaching Assistant, NC State University 4 semesters

- Fundamentals of Microeconomics (Master-Level): Fall 2024

- Intermediate Microeconomics: Fall 2020, Spring 2021
- Intermediate Econometrcis (Master-Level): Spring 2025

Lab Instructor, NC State University — 2 semesters

- Principles of Macroeconomics: Fall 2021, Spring 2022

Fellowships, Awards and Grants

Jon & Kathryn Bartley Scholarship, NC State University — 2020

Owens Graduate Fellowship, NC State University — 2021

Toussaint Scholarship, NC State University — 2022

Goins Economic Graduate Education Scholarship, NC State University — 2022

Andrew & Thelma Scholarship — 2023

Econ Graduate Fellowship, NC State University — 2023, 2024

Conference Presentations

World Finance Conference, Ireland — 2026

Southwestern Finance Association, San Antonio — 2025

European Winter Meeting of the Econometric Society, Spain — 2024

Asia Meeting of the Econometric Society,East&Southeast Asia conference, Vietnam — 2024

CES China Conference, Hangzhou — 2024

Agricurical & Applied Economics Association(AAEA) Annual Meeting (Poster Session),New Orleans — 2024

NC State University BrownBag Student Seminar, Raleigh — 2023,2024

Other Information

Programming: Python, R, C++, STATA, MATLAB, Rust, Java, Julia, MySQL

Language: English(fluent), Chinese(native)

References

Prof. Mehmet Caner (Advisor)
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Prof. Denis Pelletier
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